

Mutendi Sambaza

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EDUCATION

Bachelors of Computer Science (Hons.)

Expected May 2027

Carleton University

Ottawa, ON

Year Standing: 4th **GPA:** 3.8/4.0

Awards: Award of Excellence for Internationals (2022), President's Scholarship (2022)

Relevant Coursework: Statistical Modeling, Data Structures and Algorithms, Computer Systems, Operating Systems, Systems Programming, Discrete Structures, Fundamentals of Web Applications, Software Engineering.

EXPERIENCE

IBM - Software Engineer, Automations and Integrations

May. 2025 – present

IBM Global Sales, TechXchange

Ottawa, ON

- Architected and automated the Champions Nomination Platform for IBM's Premium Client Advocate tier, applying AI classification and a logic-driven scoring system to handle over **20,000+ applications**, cutting review cycles by **80%**.
- Developed the Champions Slack Management Tool, a CLI automation using **Selenium** that replaced a manual 5-month workflow with a one-click solution, improving productivity by **98%** and saving **\$50K+ in annual operational costs**.
- Engineered a secure, offline-first Transcription Tool using `whisper.cpp` and `ffmpeg` to enable concurrent transcription in air-gapped environments, accelerating batch processing and debugging.
- Pioneered additional scalable automation initiatives to abstract **80%+** of manual data handling.
- Orchestrated a cross-functional team in the WatsonX Challenge, designing "AgentX," a modular AI framework for natural-language creation of specialized roles.

SKILLS

Languages: Java, C/C++, Python, JavaScript, SQL (MySQL)

Frameworks & Databases: React, Tailwind CSS, Flask, Node.js, ChromaDB, MediaPipe

DevOps: AWS, IBM Cloud, Docker, Linux, Bash, Git, CI/CD.

Libraries: Selenium, NumPy, JavaFX, OpenAI API, OpenCV, GDB, Valgrind

PROJECTS

Lokdin | *Python, React, TypeScript, OpenCV, MediaPipe, Flask, TensorFlow, Keras, NumPy*

- Led a team to secure **3rd place overall at the CUHacking Hackathon**, building a custom AI-powered engagement detection tool with applications in education, healthcare, and corporate workflows.
- Engineered **feature extraction pipelines**, implementing advanced algorithms for facial landmark detection, gaze tracking, and engagement classification, achieving **97% accuracy** on synthetic datasets for real-time insights.
- Developed **visualizations and heatmaps** to translate engagement data into user-friendly feedback, enabling users to derive actionable insights **30% faster**

Noted | *React, JavaScript, Tailwind CSS, Python, OpenAI, ChromaDB, Flask, Docker*

- Developed a **full-stack transcription platform** to enhance accessibility of educational content for students featuring annotatable transcripts and PDF downloads for offline study
- Integrated OpenAI Whisper and YouTube API for automated, accurate transcriptions, with ChromaDB storing vector embeddings to enable **Retrieval-Augmented Generation (RAG)** for contextually relevant summaries.
- Built a robust Flask backend to manage API calls, handle user sessions, and ensure efficient data management, alongside a responsive UI.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

ColorStack Member

Jan. 2025 – Present

Founder and President

Mar. 2024 – Present

Black Students in Tech Nexus

- Launched and grew membership to **30+ students**, organizing networking events with industry professionals while facilitating professional skill building.
- Secured contracts for **web development projects**, leading a team of developers and designers to build and deploy websites for small organizations and local businesses, supporting their digital presence and community impact.